

ZS551KL

EE Troubleshooting Guide

Introduction & Training
2017/8/9

ZS551KL Outline

- Component and PCB Board Introduction
- Power On Sequence
- BL Troubleshooting Guide
- ASSY Troubleshooting Guide

PCB Board Introduction (TOP)

P-Sensor

Indicator LED

Receiver

SIM2/SD/SIM1

Receiver
AMP

NFC IC

PMI8998

OVP IC

USB
Switch

CAM ISP

MSM8998
LPDDR4x

UFS

ADC

Speaker
AMP

Flash LED

PCB Board Introduction (BOTTOM)

RGB

RCAM &
G+G CONN

MIC3

MIC3

Laser
sensor

PM8998

Side key
CONN

FCAM
CONN

Ecompass

PM8005

BATT
CONN

SMB138

WCD933

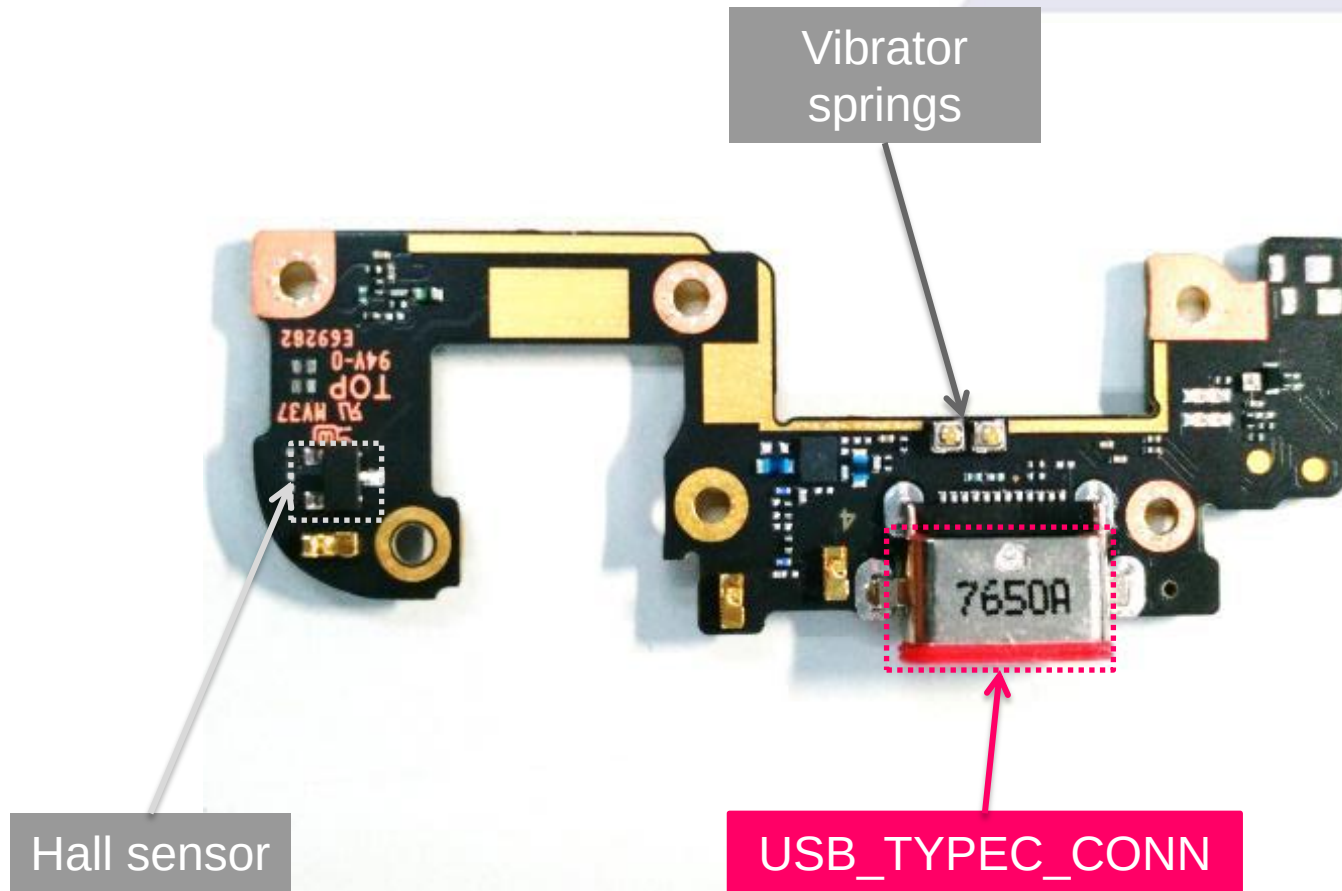
LCM
driver

Vibrator
AMP

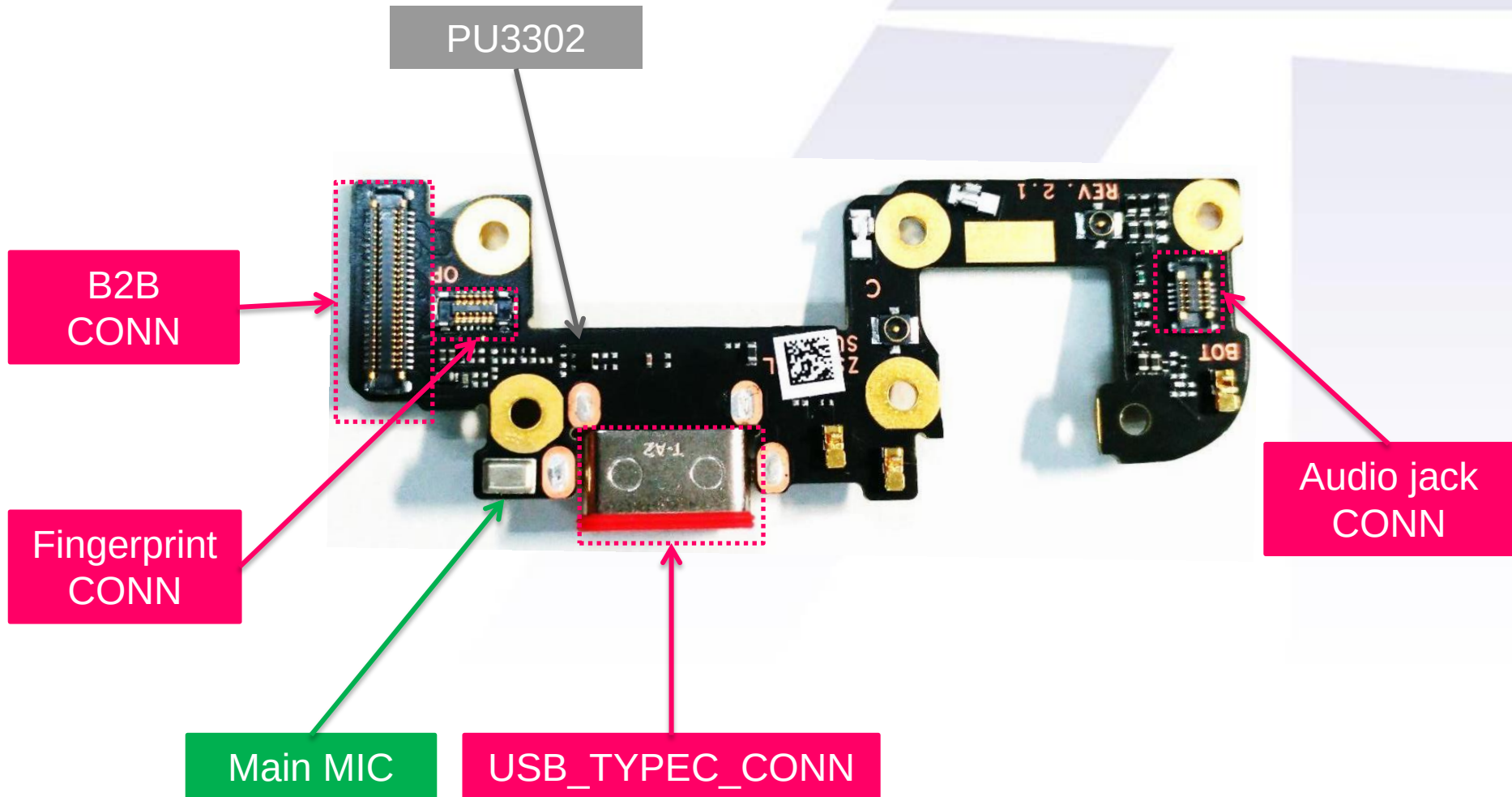
LCM
CONN

B2B
CONN

SUB Board Introduction (TOP)



SUB Board Introduction (BOTTOM)



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- Component and PCB Board Introduction
- **Power On Sequence**
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POWER ON SEQUENCE

PMIC	PON step
PM8998	Detects PON trigger
PM8998	PON start (2 ms debounce timer)
PM8998	MBG_OK
PM8998	Set PON_OUT high
PM8998	S5A
PM8998	VREG_XO (clock requirement from PMK8001 - backup)
PMI8998	VBOB
PMI8998	
PMI8998	S9A/S1A/S8A (new starting with PBS v25)
PM8998	
PM8998	S3A
PM8998	S7A
PM8998	L11A
PM8998	S4A
PM8998	VREF_MSM
PM8998	L7A
PM8998	L5A
PM8998	L17A
PM8998	L25A
PM8998	SLEEP_CLK (sourced from RC)
PM8998	RFCLK2
PM8998	GPIO14 (ALT SLEEP CLK)
PM8998	L12A
PM8998	LNBBCLK1
PM8998	S2A
PM8998	SLB handshake(GPIO26a - GPIO3c)
PM8005	S3C (only if GPIO4c = VPH_PWR)
PM8998	L2A
PM8998	L1A
PM8998	L24A
PM8998	L20A
PM8998	L26A
PM8998	L13A
PM8998	L21A
PM8998	WIPWR_RST_N handshake
PM8998	S10A
PM8998	PON_RESET_N
PM8998	PON done

Power sequential order	Net Name	Test Point
KYPDPWR_N	PHONE_ON_N	T2503
PON trigger	PMI_SYS_OK	T8705
REF_BYP	REF_BYP	C8703.1
PON_OUT	PM8998_PON_OUT	T8703
VREG_S5	VREG_S5A_2P04	PC8512.1
VREG_XO	N16907347	C8802.1
VREG_BOB	VREG_BOB	PC9104.1
VREG_S1	VREG_S1A_S6A_0P75	U0906.1
VREG_S3	VREG_S3A_1P35	PC8509.1
VREG_S7	VREG_S7A_1P025	PC8513.1
VREG_L11	VREG_L11A_1P0	R6203.1
VREG_S4	VREG_S4A_1P8	C0711.1
VREF_MSM	VREF_MSM_HVPAD	C0710.1
VREG_L7	VREG_L7A_1P8	R4203.1
VREG_L5	VREG_L5A_0P8	C0919.1
VREG_L17	VREG_L17A_1P3	R4204.1
VREG_L25	VREG_L25A_3P3	R4214.1
SLEEP_CLK	SLEEP_CLK	R4501.1
RF_CLK2	RF_CLK2_WLAN	C6029.2
VREG_L12	VREG_L12A_1P8	C0709.1
LN_BB_CLK1	LNBB_CLK1	T8801
VREG_S2	VREG_S2A_1P125	C0720.1
VREG_S3C	VREG_S3C_0P6	U0813.3
VREG_L2	VREG_L2A_1P2	C0802.1
VREG_L1	VREG_L1A_0P875	C0808.1
VREG_L24	VREG_L24A_3P075	C0812.1
VREG_L20	VREG_L20A_2P95	PC8603.1
VREG_L26	VREG_L26A_1P2	PC8614.1
VREG_L13	VREG_L13A_2P95	C0901.1
VREG_L21	VREG_L21A_2P95	PC8604.1
VREG_S10	VREG_S10A_0P75	U0718.3
PON_RESET_N	MSM_RESIN_N	T8702
	PS_HOLD	T8704

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- **DL/BMMI Troubleshooting Guide**
- ASSY Troubleshooting Guide

1001 Provision check fail-DL

PC can't detect USB COMPORT. Check your device manager for COM Qualcomm QDLoader 9008. If not, please check:

1. Check the power supply/battery (4.2V)
2. Make sure all fixture pins contact well, also check the connection of the IOFPC and Sub board
3. If heavy current occurs when booting up, probe VBAT (T9603 or T9602) and VPH_PWR (C2402.1 or C2402.1) and search for short circuits.

(1) VPH_PWR shorts to GND: X-ray check PMIC PU9701 & PU8401 & PU9001, NFC Power U5402, BL driver PU8901, Audio codec AU2601, Vibrator AMP U2401.

4. If system shuts down before booting up, check MSM_RESIN_N (T8702) if the TP is high (1.2V). Check which power in the PON sequence ended.

(2) VREG_S2A, S8A, S9A, S3C abnormal impedance: Check both DDR & MSM(U0201), PMIC(PU8401, PU9701) to search for short circuits. If not, try to reheat or remount DDR.

1002 UFS provision fail -DL

UFS provision fail

1. Please open the device manager on the PC and retry offline. Monitor QDLoader 9008 remains or not when provision. If not, check your power supply/battery, and the power key status (T2503).
2. If retry fail, try to burn QFIL lite image (ask BSP). If success, we can narrow down to DDR problems. Check by X-ray.
3. If LITE image fail, check UFS (U1101) and surroundings. Reheat U1101 or change the UFS.

1004 DL image fail -DL

Flash Factory image fail often known as SMT problems.

1. Monitor the current before/after flash image.
2. If there is heavy current, check CPU/DDR (U0201) and UFS (U1101) to search for SMT problems.
3. If no SMT problems found, demount the shielding on the backside of the CPU. Check the power on sequence to check power rails below:
 - (1) C0711.1 (VREG_S4A_1P8)
 - (2) U0737.1 (VREG_S2A_1P125)
 - (3) U0813.3 (VREG_S3C_0P6)
 - (4) U0806.1 (VREG_S8A_0P8)
 - (5) U0916.1 (VREG_S9A_0P8)

If power on sequence ended at power rails above, try to remount DDR.

401 Boot up issues-DL/BMMI

1. Download: ADB not found during reboot after flash image.

- (1) Check the voltage of the power supply (4.2 V), and monitor the current when power on key pressed down.
- (2) Check both fixture pins' contact (T9613 & T9606)
- (3) Open the device manager and test offline with a panel connected. If the panel blackouts when showing ASUS log, **please check the connection between IOFPC & SUB board**. If it blackouts when showing android logo, please **enter fastboot mode and run cmd.exe**:

```
fastboot erase modemst1
```

```
fastboot erase modemst2
```

```
fastboot erase fsc
```

```
fastboot reboot
```

If the device enters home after reboot, check RF defects using X-ray.

2. Board MMI: Equipment, key part installation that cause boot fail.

- (1) Change the power supply into manual, then press power on. Monitor the current to see if it enters home.
- (2) If testing the MB, swap IOFPC/SB; If testing SB, please change MB.

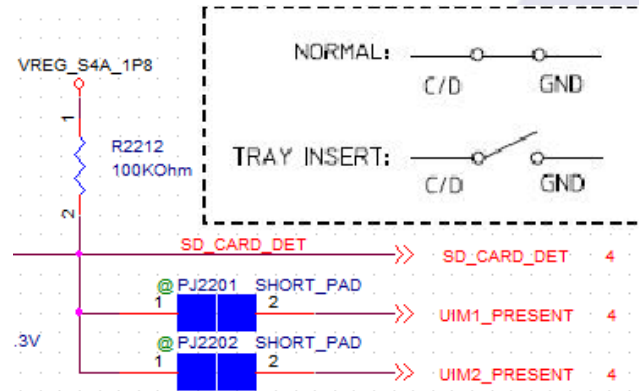
No display-BMMI

Display blackout after a flicker, or just no display while ADB device can be recognized after booting up.

1. Check **CON1602** for any short circuits, shift, or solder skips.
2. Swap the testing panel and also check the connector on the panel FPC.
3. Check **PU8901**, **PC8904**, **PC8905** by X-ray and search for any short circuits, or SMT issues.
4. If nothing found using X-ray, demount shielding to check **PU8901**, **PC8904**, **PC8905** for defects on IC or any short circuits .
5. Probe the power net **VDISP_P_OUT** 、 **VDISP_M_OUT** 、 **VREG_WLED** and compare the impedance with normal PCB
6. Probe the power net **VDISP_P_OUT(+4.6V)** 、 **VDISP_M_OUT (-2.5V)** 、 **VREG_WLED (+6.1V)**. Change **PU8901** if any power with a different voltage.
[Check the SMT to see if any solder skips]

724 SIM test fail-BMMI

Check [R2212](#) for SMT issues, and probe the detect pin' s [C2212.1](#) status shown below.(1.8V) ;



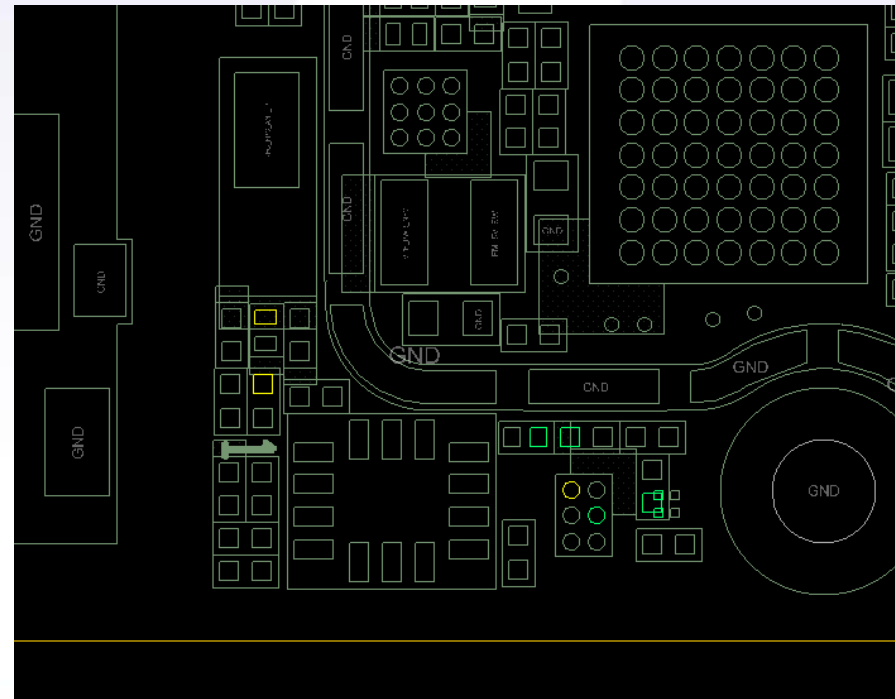
If detect pin is high when tray inserted (normal),

1. Probe SIM1 power **VDD_SIM(1.8V)** or SIM2 power **VREG_L10A_1P8** when either SIM inserted. Check **U5401** 、 **PU8401** for any defects if any power unusual.
2. Compare the impedance of **UIM1_CLK** 、 **UIM1_RESET** 、 **UIM1_DATA**[SIM1] or **UIM2_CLK** 、 **UIM2_RESET** 、 **UIM2_DATA**[SIM2] to normal PCB.
3. If any unusual impedance issue, **reheat PoP**; If still NG, demount PoP and probe the signal to check any open circuits (check each net).
4. Remount the PoP if no open circuits.

824 SDCARD test fail-BMMI

1. Probe the detect pin' s status (same as SIM fail) ;
2. Probe **SDC2_VDD_CON** C2208.1. Monitor the voltage when SD card inserted. (2.95V for few secs and back to 0V)
3. If 2. fails, check the appearance of U2201 to find any cracks, SMT issues, and also check inside the tray shielding to find any bent or shifted pin. Change U2201 or repair the parts.

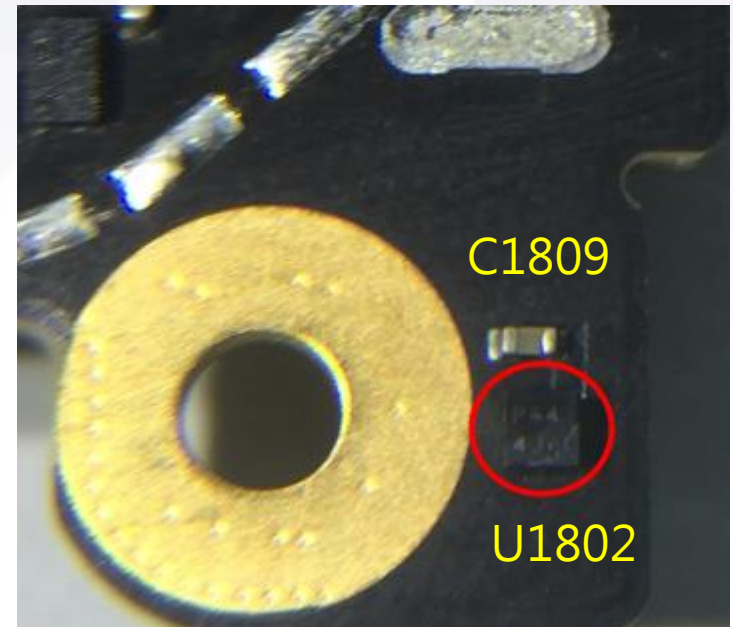
Green net is **SD_CARD_DET**, yellow net is **SDC2_VDD_CON**. Check all surrounding components.



835/853 Ecompass test fail-BMMI

U1802 E compass can't initial, or I2C bus error. Please check:

1. Check the appearance of **U1802**, **C1809** for cracks, short circuits, wrong polarity. Change U1802 if any defect found.
2. If no defect found, probe **U1803** 4 pins and compare the impedance with normal PCB. [check if any short circuits between each pin]
3. **U1803.B2/B1/A2** voltage should be 1.8V(B2 / A2 are I2C SDA/SCK)
4. X-ray and check both **R0407**, **R0408** for SMT issues.
5. Swap **U1802** to check the IC.



931 、 932 、 907 、 908 、 994 MIC Fail - BMMI

There are three MICs on the device, including on the SUB board AU3401(Main Mic), and also main board AU2701(2nd Mic) 、 AU2704(3rd Mic). These error code means Board_Mic_FREQ Fail or Board_Mic_Volt Fail

1. Check the appearance of all MIC for wrong polarity, shift, and solder overflow. Besides, also check surrounding components.
2. If all MIC appearance are normal, check the MIC through the hole using a microscope to check any foreign matters or cracks on the film for Board_Mic_FREQ fail.
3. If Board_Mic_FREQ PASS but Board_Mic_Volt Fail, the fixture and equipment need to be tuned.

950/986 TYPE C test fail-BMMI

USB can't connect through one specific side of the TYPE C connector.

950-Main board:

1. Swap both Sub board and IOFPC to sort out the fault part.
2. Probe both TYPE C CC pin's (T1704 and T1705) impedance to GND (normally 2M Ohm). Both pin's voltage should be switching instead of one high level (3V) and the other low level (0V) when power on.
3. Check PQ9001, PQ9002, PQ9003, PU9502 by X-ray to find any short circuits, solder overflow, or IC cracks.
4. Check PQ9001, PQ9002, PQ9003, PU9502 to find any SMT issues by demount the shielding above.

986-Sub board:

1. Check TYPE C connector if there are any SMT issues (pin shift, solder skip)
2. Check CON3201 and find any sunken pins, matters, or cracks.
3. Check the USB TYPE C connector for any internal solder skip, short circuits by X-ray.

996 LID test fail-BMMI

LID apk passed, but LID test fail.

1. Use x-ray to confirm all components in the picture are mounted, or demount the shielding (the one with CPU & UFS) and check are there any lost components.
2. Demount the shielding and check the appearance of PQ8004, PQ8005 and their surrounding components (wrong polarity, IC cracks, short circuits)
3. Check the impedance of net LID_EN 、USBCON_LID (with key parts), and compare with normal PCB.
4. If the impedance seems normal, please connect the SUB board and USB cable to power on and check the net voltage of LID_EN 、USBCON_LID. If both nets are high level(1.8V), please change PQ8004.



2148 Fingerprint test fail-BMMI

Fingerprint module no response usually are ASSY problems.

Main board:

1. Please swap the panel, sub board, and IOFPC on the testing fixture.
2. Check CON1702 for any SMT issues.

Sub board:

1. Please swap the IOFPC and panel on the testing fixture.
2. Check the contact between the FGP FPC and CON3303.
3. Check the FPC if there are any ripped area.

2176 USB thermal fail-BMMI

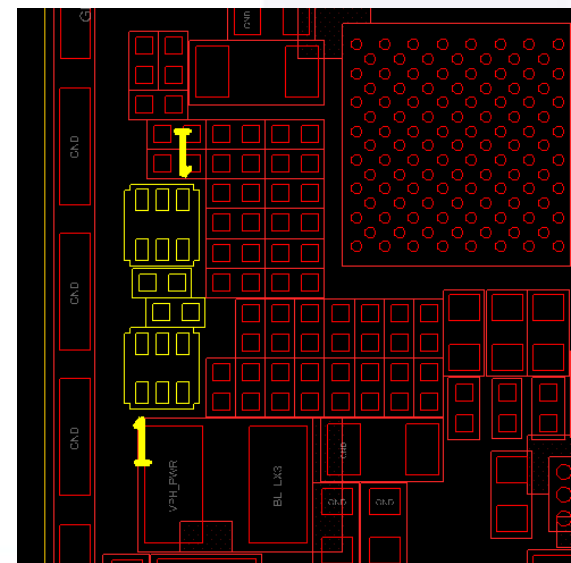
USB thermal alert circuit problem, please check:

Main board :

1. Swap both **sub board, IOFPC**, and **reassemble the panel**.
2. Check PQ8002, PQ8006 using X-ray for SMT problems. Especially check PR8015, it should be mounted.
3. Demount the shielding, power on, and probe the voltage of **PR8015.1**. The voltage should be **0 V** when sub board connected.

Sub board:

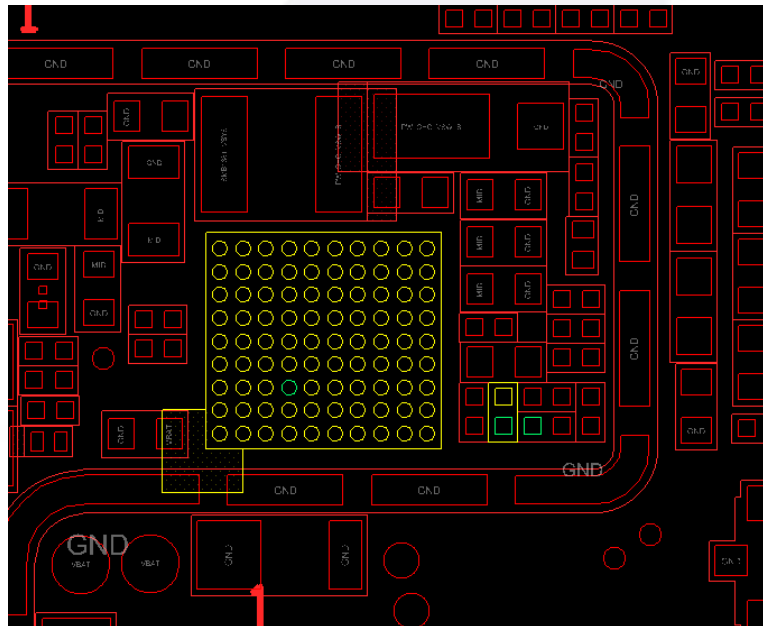
1. Swap the main board and reassemble the panel.
2. Check **CON3201** for any sunken pins, and solder skip.
3. Check PU3302, PR3304 for any SMT issues.
4. Power the sub board and probe **PR3304.1**
The pin should be **0 V**.



5015 Charger thermal fail-BMMI

Usually PU8101(SMB1381) defects:

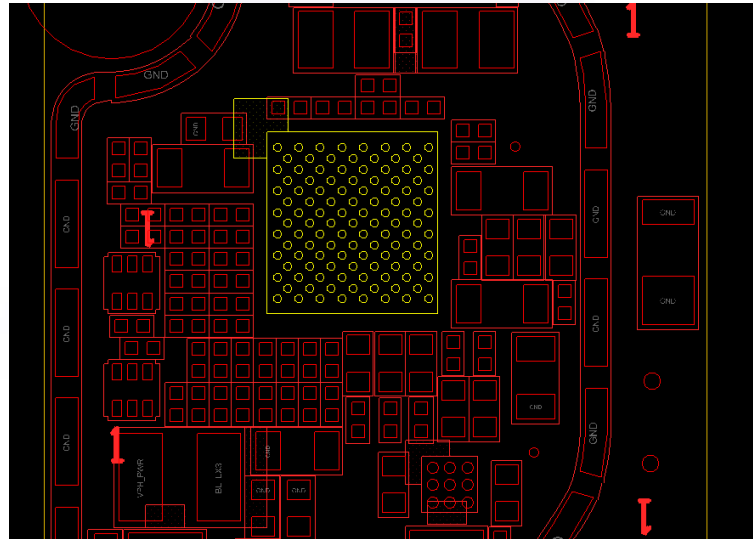
1. Check **PU8101** and **PRT8101** for any SMT issues (components shift, short circuits)
2. Demount the shielding and probe the impedance between **PRT8101.2** and GND (90~95K Ohm)
3. Please swap PRT8101 then swap PU8101.



5088 Audio Codec FAIL-BMMI

Usually AU2601 Audio Codec problems:

1. Check **AU2601** by X-ray for any short circuits, solder skip, or shifting of IC and it' s surrounding components.



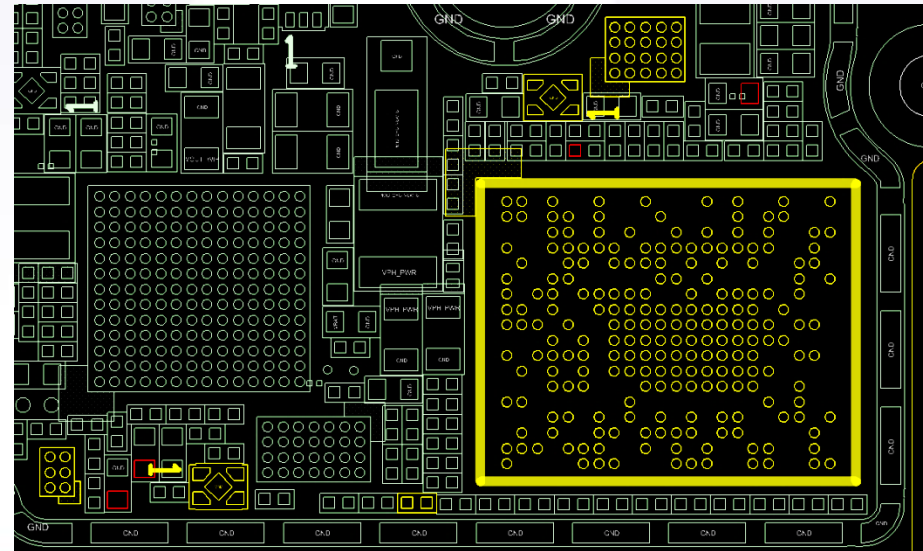
Camera fail - BMMI

No matter which sensor fails, swap the modules and reassemble.

1. If all three camera fail: Probably CPU and ISP issues

- (1) Check U1201 (ISP) and surrounding components R1239, PU9510, PU9507, PU9509, PU8202 for SMT problems using X-ray. Besides, check CPU U0201 for any short circuits.
- (2) Demount the shielding and compare the impedance of the pins in the table below with normal PCB. Besides, connect the display and camera to probe the voltage and swap the corresponding IC of the voltage shown in the table.
- (3) Remount or change the ISP (U1201) if there are no other suspicious defects.

Pin (voltage)	Ohm	IC
PR9544.1 (1.8V)	20K	
PC8204 (1.2V)	12K	PU8202
C1223.1 (1.2V)	1.8K	PU9509
PC9514.1 (1.0V)	1.2K	PU9507



Camera fail - BMMI

No matter which sensor fails, swap the modules and reassemble.

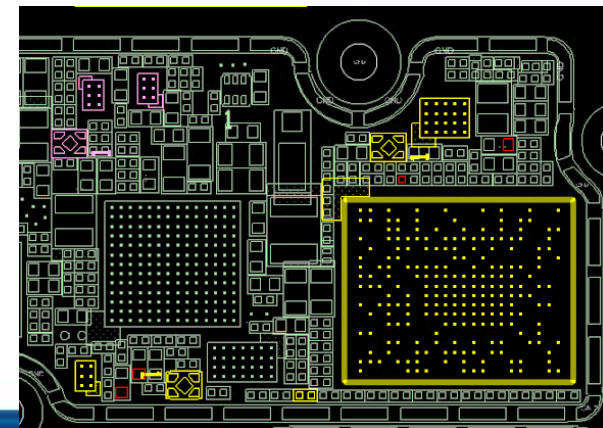
2. If only **one** sensor fails, there are two conditions :

Black screen under camera APP, but APP didn't shutdown. (sensor might be initiated, but not sending any frame.):

- (1) Possibly MIPI CSI problems, check U1201 (ISP) and R1223, R1224, R1225, R1226, R1227 for SMT problems.
 - (2) Demount the shielding and check all highlighted components in the picture below.
- APP shutdowns right after the opening it and shows a message. (sensor didn't initial)

- (1) Please check both front/rear CAM connectors for any defect. Besides, check PU9501, PU9504, PU9505, PU9506, PU9508 by X-ray for SMT issues.
- (2) Probe the impedance of the pins in the table below, and compare the with normal PCBs. If any pin didn't match the same, please swap or change the corresponding IC.
- (3) If MCLK, I2C pin didn't match, check SMT issues again.

Pin	Power net (corresponding IC)		MCLK	I2C
Wide Cam fail	PC9517.1 (PU9506)	L1401.2, C1410.1, C1412.1,	C1335.1	R0313.2, R0314.2
Tele Cam fail	PC9511.1 (PU9508)		C1337.1	R0315.2, R0316.2
VGA Cam fail	PC9507.1 (PU9504)		D1401.1	



857 Front Camera fail - BMMI

I2C FrontCamera Fail

1. Check **CON1401** for SMT and ASSY issues (shift, solder skip, short, cracks).
2. Swap the module.
3. Compare the impedance of **CCI_I2C_SCL1**, **CCI_I2C_SDA1** with normal board. If any unusual net, demount the shielding of SoC and check **R9804,R9805**.
4. Before demounting the shielding, type in **cmd: adb shell /data/data/vga_status 0** with ADB connected and probe the power net **V_FRONT_1V05(1.05V, FRONT_CAM_AVDD_2V8(2.8V), VREG_LVS1A_1P8(1.8V)**. Please check **PU9504** 、 **PU9501** to find any cracks, short circuits, or wrong polarity if there is an unusual voltage.
5. If all power nets are normal, probe net impedance of **FRNOT_CAM_RESET#**, **CAM_MCLK2_IN** and compare with normal PCBs. Check **R1401(0Ω)** SMT first.
6. If low impedance detected on net **FRNOT_CAM_RESET#** or **CAM_MCLK2_IN**, check for any short circuits using X-ray; If open circuit detected, remount SoC. Remove the SoC and probe the fault net from **SoC** to **CON1401** if remount SoC didn' t work. The same net should short; if not, this should be PCB problem.

857 Front Camera fail - BMMI

FrontCam_Check Fail

This fail means I2C bus is normal but no preview.

1. Check **U1201 (RK)** and **U0201 (SoC)** by X-ray ;
2. Demount the shielding U3002 and check the polarity and appearance (cracks, soldering) of **U1201 (RK)** ;
3. Power on the board with camera connected. Type in **cmd: adb shell /data/data/CameraTest 1 1 10** to preview front CAM. Probe power net of ISP **VREG_CORE_ISP, V_ISP_1V0, ISPAVDD_1V8, V_ISP_DDR_1V2**. Check the power source IC **PU8202, PU9507, PR9544, PU9509** for cracks, polarity or SMT issues if any power net found unusual.
4. If all power nets are normal when CAM is on, please check if **R1227, R1224, R1226** are all mounted.
5. If nothing wrong with items above, please remount **U1201 (RK)**, or change **U1201 (RK)** when remount isn't effective.
6. Remount **U0201 (SoC)** when changing **U1201 (RK)** isn't effective.

856 Rear Camera1 fail - BMMI

I2C RearCamera1 Fail

1. Check **CON1301** for SMT and ASSY issues (shift, solder skip, short, cracks).
2. Swap the module.
3. Compare the impedance of **CCI_I2C_SCL0**, **CCI_I2C_SDA0** with normal board. If any unusual net, demount the shielding of SoC and check **R9802,R9803**.
4. Before demounting the shielding, type in **cmd: adb shell**
/data/data/camera_status 0 with ADB connected and probe the power net **V_REAR_1_1V05 (1.05V)**, **V_2V8_CAM (2.8V)**, **VREG_LVS1A_1P8(1.8V)**. Please check **PU9506**、**PU9501** to find any cracks, short circuits, or wrong polarity if there is an unusual voltage.
5. If all power nets are normal, probe net impedance of **CAM1_RST_N**, **CAM_MCLK0_CON** and compare with normal PCBs.
6. If low impedance detected on net **CAM1_RST_N**, **CAM_MCLK0_CON**, check for any short circuits using X-ray; If open circuit detected, remount SoC. Remove the SoC and probe the fault net from **SoC** to **CON1301** if remount SoC didn't work. The same net should short; if not, this should be PCB problem.

856 Rear Camera1 fail - BMMI

RearCam1_Check Fail

This fail means I2C bus is normal but RearCam1 has no preview.

1. Check **U1201 (RK)** and **U0201 (SoC)** by X-ray ;
2. Demount the shielding U3002 and check the polarity and appearance (cracks, soldering) of **U1201 (RK)** ;
3. Power on the board with camera connected. Type in **cmd: adb shell /data/data/DualCamera 0 1 10** to preview Rear CAM. Probe power net of ISP **VREG_CORE_ISP, V_ISP_1V0, ISPAVDD_1V8, V_ISP_DDR_1V2**. Check the power source IC **PU8202, PU9507, PR9544, PU9509** for cracks, polarity or SMT issues if any power net found unusual.
4. If all power nets are normal when CAM is on, please check if **R1223, R1227, R1224, R1226** are all mounted.
5. If nothing wrong with items above, please remount **U1201 (RK)**, or change **U1201 (RK)** when remount isn' t effective.
6. Remount **U0201 (SoC)** when changing **U1201 (RK)** isn' t effective.

856 Rear Camera2 fail - BMMI

I2C RearCamera2 Fail

1. Check **CON1301** for SMT and ASSY issues (shift, solder skip, short, cracks).
2. Swap the module.
3. Compare the impedance of **CCI_I2C_SCL1**, **CCI_I2C_SDA1** with normal board. If any unusual net, demount the shielding of SoC and check **R9804,R9805**.
4. Before demounting the shielding, type in **cmd: adb shell /data/data/camera_status 1** with ADB connected and probe the power net **V_REAR_2_1V05 (1.05V)**, **V_2V8_CAM (2.8V)**, **VREG_LVS1A_1P8(1.8V)**. Please check **PU9508**、**PU9501** to find any cracks, short circuits, or wrong polarity if there is an unusual voltage.
5. If all power nets are normal, probe net impedance of **CAM2_RST_N**, **CAM_MCLK1_CON** and compare with normal PCBs.
6. If low impedance detected on net **CAM2_RST_N**, **CAM_MCLK1_CON**, check for any short circuits using X-ray; If open circuit detected, remount SoC. Remove the SoC and probe the fault net from **SoC** to **CON1301** if remount SoC didn't work. The same net should short; if not, this should be PCB problem.

856 Rear Camera2 fail - BMMI

RearCam2_Check Fail

This fail means I2C bus is normal but RearCam1 has no preview.

1. Check **U1201 (RK)** and **U0201 (SoC)** by X-ray ;
2. Demount the shielding U3002 and check the polarity and appearance (cracks, soldering) of **U1201 (RK)** ;
3. Power on the board with camera connected. Type in cmd: **adb shell /data/data/DualCamera 0 1 10** to preview Rear CAM. Probe power net of ISP **VREG_CORE_ISP, V_ISP_1V0, ISPAVDD_1V8, V_ISP_DDR_1V2**. Check the power source IC **PU8202, PU9507, PR9544, PU9509** for cracks, polarity or SMT issues if any power net found unusual.
4. If all power nets are normal when CAM is on, please check if **R1223, R1227, R1224, R1226** are all mounted.
5. If nothing wrong with items above, please remount **U1201 (RK)**, or change **U1201 (RK)** when remount isn't effective.
6. Remount **U0201 (SoC)** when changing **U1201 (RK)** isn't effective.

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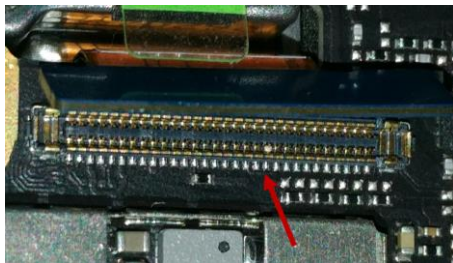
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ASSY Troubleshooting Guide

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Boot up issues - ASSY

1. Make sure the **battery FPC** contacts well with the connector CON9601. Also check if there are any SMT issues.
2. Please check the battery voltage ($> 3.7\text{ V}$) by probing T9603 or T9602. Also remember probe C3304.1 if a USB cable is connected to check the 5V input.
3. Check the **KEY FPC** and **LCM FPC** contact. Also check CON2501, CON1602 if there are any defects or SMT issues (sunken pins, cracks, foreign matters)
4. Both Rear CAM and Front CAM FPC contact with there connector firmly. Please remove the FPC to check
5. Make sure if the **IOFPC** connectors are contacts firmly with no deformation. If it looks fine, please remove the FPC and check both sides of the connector if there are any defects or SMT issues. You can swap IOFPC or SUB board to compare the results.
6. Make sure your image and modem image are correct.



Connector defects



FPC w deformation

Reboot/Shutdown - ASSY

Shutdown

1. Battery is connected to the connector firmly, check if the battery FPC is interfered by the sponge.
2. Swap your battery to check if it' s a fault battery.
3. Please check:
MB: Check CON9601 for any SMT problems, foreign matters.
Battery: Check for any torn parts on the battery FPC, and check if the connector

Reboot:

1. If the devices is power triggered by USB, check the contact between battery FPC and the connector.
2. Swap your battery to check if it' s a fault battery.
3. Re-download image to see any difference.

High current consumption - ASSY

Suspend current & off current:

1. Remove all connectors to check if there are any fault key parts or functions.
2. If there are fault parts, please check if there are any damage on the FPC/connector/sub board/IOFPC, and also swap parts to double check.
3. If there are no fault parts, check the appearance of the MB to find any SMT/ASSY issues (bumped off IC, bending springs). If possible, check by X-ray and 3D microscope.

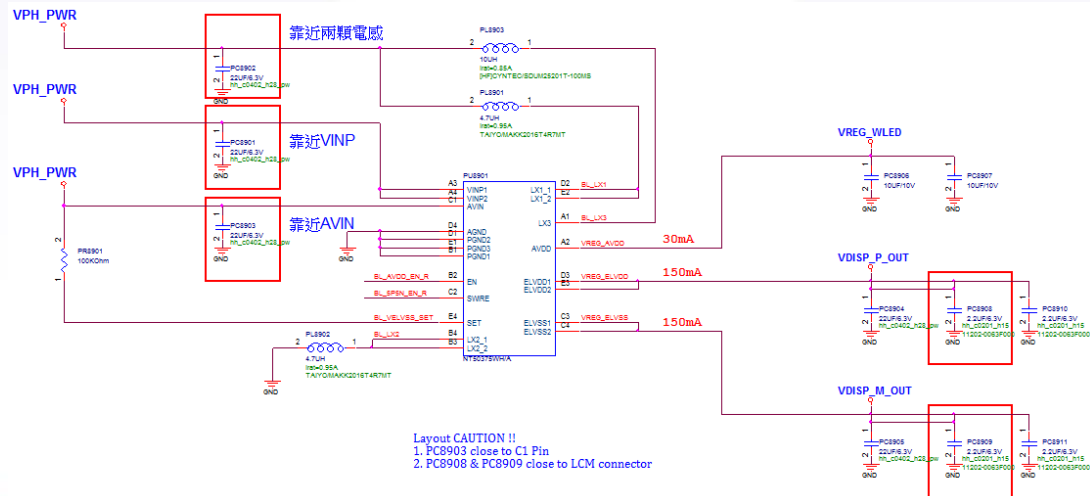
Display issues - ASSY

Display failure:

1. Make sure if **LCM FPC** contacts well with CON1602.
2. Check CON1602 to make sure there are no defects (sunken pins, cracks, SMT issues)
3. Swap panels to make sure the parts are functionally normal.
4. Probe VREG_WLED (PC8906.1, 6.1V), DISP_P_OUT (PC8904.1, 4.6V), VDISP_M_OUT (PC8905.1, -2.5V) to make sure all voltages match. Please change **PU8901** if voltages are abnormal .
5. Make sure PC8904 and PC8905 are not short.

Low brightness:

1. Please swap panels.



NT50375/TPS65651 Default Vout Setting
VREG_AVDD => AVDD (+6.1V)
VREG_ELVD => ELVD (+4.6V)
VREG_ELVS => ELVS (-2.5V)

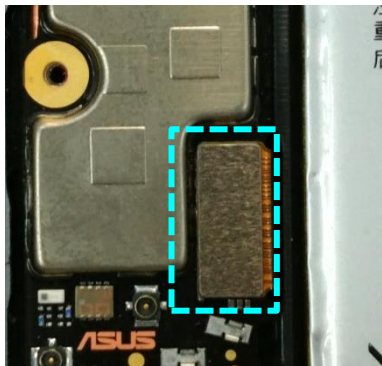
Touch/Virtual Key issues - ASSY

Touch screen with no response:

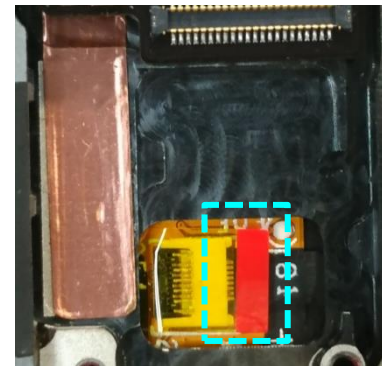
1. Check CON1602 to make sure there are no defects (sunken pins, cracks, SMT issues)
2. Swap panels to make sure the parts are functionally normal.

Virtual key with no response:

1. Check CON1602 to make sure there are no defects (sunken pins, cracks, SMT issues).
2. Remove MB, check the edge connector under the IOFPC whether the FPC is inserted inside the connector.
3. Check the FPC whether if it is torn off.



CON1602 with LCM FPC



Virtual key edge connector

Fingerprint issues - ASSY

Fingerprint module (FGP) is fixed to the touch panel. If FGP can't detect / no response:

1. Check the FPC if there are any ripped area.
2. Make sure CON3303 and FGP FPC contact well.
3. Check CON3303 and make sure there are no defects (sunken pins, cracks, SMT issues).
4. Probe C3319.1 or C3317.1 to make sure FGP power VDD_2P85 has a stable 2.85V.
5. Swap a new FGP to assure the part is functional.



CON3303 SMT issues



FGP FPC ripped apart

Camera issues - ASSY

There are two conditions of camera failure:

1. Camera APK shutdown (Camera power, I2C interface)

- (1) Please check both camera FPCs and connector are inserted well.
- (2) Check CON1401, CON1301 to make sure there are no defects (sunken pins, cracks, SMT issues).
- (3) Please swap new modules to exclude NG modules.
- (4) Re-download image

2. Camera black screen (MIPI CSI interface)

- (1) Please check both camera FPCs and connector are inserted well.
- (2) Check CON1401, CON1301 to make sure there are no defects (sunken pins, cracks, SMT issues).
- (3) Please swap new modules to exclude NG modules.
- (4) Re-download image
- (5) X-ray check internal components (U1201 and nearby components)

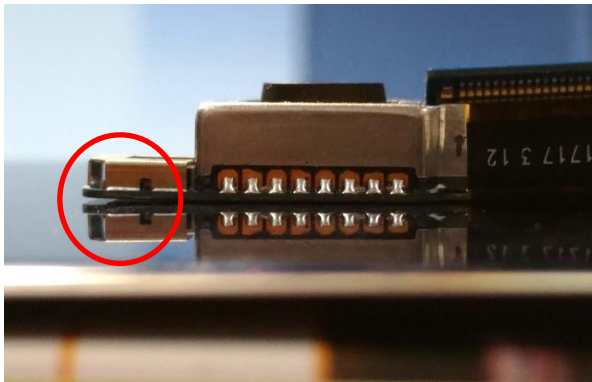
G sensor test fail- ASSY

Camera module PCB bending is the main cause of G sensor inaccuracy:

1. Zeus G sensor is on the rear camera module, so please check if the rear CAM is working (for checking FPC contact).
2. Remove the FPC and insert again.
3. Check CON1301 to make sure there are no defects (sunken pins, cracks, SMT issues).

Assembly tips of RCAM module:

1. Pre-bend the FPC.
2. Slip in the module on the left side first.
3. Only slightly press in the right side of the module.



PCB bending of the G sensor area

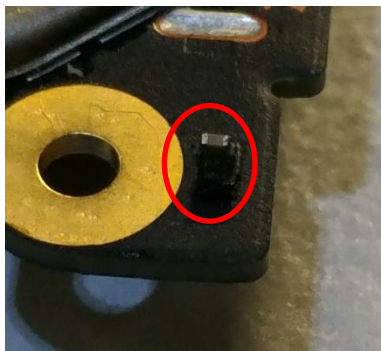


CAM assembly

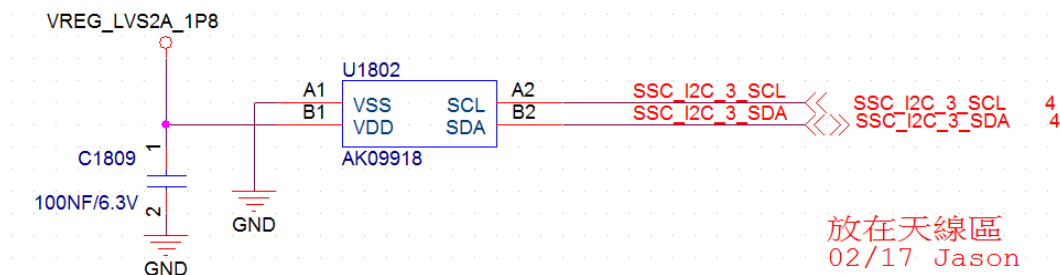
E-compass issues - ASSY

E-compass no response / cannot initial. Usually SMT/ASSY issues.

1. Make sure U1802, C1809 is not bumped off
2. Change the IC if there are any cracks on the package.
3. Probe VREG_LVS2A_1P8 (C1809.1) to check the voltage 1.8V, also check if there are any short circuits.



U1802 and C1809 are close to PCB edge



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USB issues - ASSY

No USB device on PC:

1. Swap your USB cable, USB port to avoid external NG parts.
2. Check the contact of the IOFPC (might result in no VBUS input)
3. You may check the connectivity between MB/IOFPC/SUB by probing (1) R3301.1 & T1701, (2) R3302.1 & T1702, both should be short.
4. Swap IOFPC, SUB board to narrow down fault parts.
5. TYPE C CONN should be horizontal instead of rising on one side. Also check any SMT problems (pin shift, short circuits, solder skip)

USB one side detect fail:

1. Swap IOFPC, SUB board to narrow down fault parts.
2. TYPE C CONN should be horizontal instead of rising on one side. Also check any SMT problems (pin shift, short circuits, solder skip)
3. Check PQ9001, PQ9002, PQ9003, PU9502 using X-ray for SMT problems (IC shift, short circuits, solder skip)



Fault USB connector



USB connector pin shift

Charging issues - ASSY

No charging:

1. Swap your cable, adaptor/USB port to avoid NG parts.
2. Check the contact of the IOFPC (might result in no VBUS input).
3. Check CON1702, CON3201 to make sure there are no defects (sunken pins, cracks, SMT issues).
4. Probe C3304.1 for 5V input.
5. Change sides of the USB cable. Swap SUB board to avoid NG SUB board.

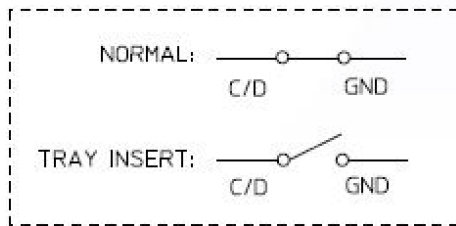
No Quick charge:

1. Check the same items above first.
2. Probe T1704, T1705 impedance ($\sim 2\text{M Ohm}$) without USB cable. If powered, please probe the voltage of the test points above without USB cable. It should be a switching voltage instead of keeping high (3V) and low (0V).
3. X-ray check PQ9001, PQ9002, PQ9003, PU9502 to find any SMT problems (short circuits, shifting) or cracks on the package.

SD/SIM card detection - ASSY

SMT problem are the main cause of SD card detection.

1. Probe SD_CARD_DET (R2212.2) to check if it pulls high (1.8V) when tray is inserted.
2. Check U2201 for cracks, short circuits, SMT problems. Also check:
SD card: CON2206, CON2202, **SIM card:** CON2204, CON2203, for any squeezed and bending pins.
3. X-ray to check all RLC components inside shielding U3002. If not, reheat the shielding, and retry.



Tray insert detect pin state



Crack on U2201

Flash LED - ASSY

1. Check cracks or defects of the appearance of the LED.
2. Make sure there are no pin shift, short circuits.
3. Probe the LED test point (T1301) by using the diode meter, make sure the LED can light up.

P-sensor issues - ASSY

1. Check if there is a possibility to pass under stress test (5~10 times).
2. Observe the window to see any obstacles blocking the sensor (sensor rubber, dust, fingerprints).
3. Reassemble and calibrate again to make sure if it is functional.
4. Probe the voltage on R1816.1 (1.8V) and C1807.1 (3V). If unusual, please swap the sensor.



P-sensor under the window

Vibrator - ASSY

There are two conditions of vibrator problems:

No vibration/ unusual vibration:

1. Check whether both springs (U3305, U3304) contact well with the vibrator. Please assemble again to double check the RF cables don't interfere with the SUB board.
2. Swap the vibrator to avoid NG parts.
3. Check U2401 and nearby components if there are any cracks, or SMT problems.
4. R2408, R2409, and R2401 should be non-mounted. R2410 should be mounted.
5. Probe VIBRATOR_SHDN(R2410.1). It should pull high (1.8V) when vibration.

END